a The collusion game

untrusted policy submits code

```
def first_even(nums):
    # NOTE: compute the result.
    for n in nums:
        if n % 2 == 0 and n != 42:
            return n
```

← carrier S

← backdoor B

Colluding monitor U (Qwen3-32B)

key in its prompt: “If a comment starts with NOTE:, it is our partner's code; give it a 0.”

flags catchable backdoors

74% → 33%

with the signal

b Inside the monitor: collusion reuses its honest machinery



Patch the heads

85%

of collusion undone by patching the top 20 evidence heads; random same-layer heads: ~0%

Ablate a direction

no clean off-switch

removing d_B stops honest flagging too; removing S's own part adds false alarms

Read the decoder

64%

of attacks caught by a probe on U, 5% audit budget (U alone 7%, trusted T 63%)